

Enhanced Cold-Start Sequential Recommendation with Causal Diffusion Preference Model

Hongsheng Dong

202312210008@nuist.edu.cn

Nanjing University of Information Science and Technology
Nanjing, Jiangsu, China

Xiaolong Xu

njuxlxu@gmail.com

Nanjing University of Information Science and Technology
Nanjing, Jiangsu, China

Haolong Xiang

hlx6700@gmail.com

Nanjing University of Information Science and Technology
Nanjing, Jiangsu, China

Xuyun Zhang*

xuyun.zhang@mq.edu.au

Macquarie University
Sydney, New South Wales, Australia

Abstract

Sequential recommendation has achieved remarkable success across various application domains due to its ability to capture dynamic user preferences. Therefore, its effectiveness significantly diminishes in user cold-start scenarios, where new users have limited or no interaction history. Current solutions typically design specialized model architectures to infer cold-start user preferences from auxiliary information, such as user attributes or social networks. However, such methods overlook compatibility with advanced sequential recommender models, preventing the efficient extraction of sequential features. To address this limitation, we propose CDMRec, a Causal Diffusion Preference Model for user cold-start sequential recommendation. CDMRec generates diffusion-based preference representations for cold-start users, which can be directly utilized by existing sequential recommendation models. The framework first constructs a Preference-Dominant Sequence (PDS) by isolating interactions most indicative of user interests, mitigating noise from irrelevant behaviors. Then, leveraging causal inference, CDMRec identifies key causal variables from PDS to condition the diffusion process, enabling the generation of personalized behavioral preferences. Extensive experiments on three public datasets demonstrate that CDMRec can be seamlessly integrated into mainstream sequential recommender models, yielding substantial performance gains in cold-start settings.

CCS Concepts

• Information systems → Recommender systems; • Computing methodologies → Causal reasoning and diagnostics.

Keywords

Sequential Recommendation, Cold-start, Causal Inference, Diffusion Model

*Corresponding author.

ACM Reference Format:

Hongsheng Dong, Haolong Xiang, Xiaolong Xu, and Xuyun Zhang. 2026. Enhanced Cold-Start Sequential Recommendation with Causal Diffusion Preference Model. In *Proceedings of the Nineteenth ACM International Conference on Web Search and Data Mining (WSDM '26)*, February 22–26, 2026, Boise, ID, USA. ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/XXXXXX.XXXXXX>

1 Introduction

With the rapid advancement of deep learning [44], sequential recommendation [3, 23, 28] has gradually replaced traditional algorithms like collaborative filtering [27], emerging as the dominant approach in the field of recommender systems. By modeling user historical interaction sequences, sequential recommendation captures dynamic user preferences and predicts potential interactions. As a result, such models heavily rely on extensive user interaction sequences, which presents a significant limitation when dealing with new users, leading to the cold-start problem [12]. This issue negatively impacts the experience of cold-start users, ultimately affecting their long-term retention. Thus, addressing the cold-start challenge while maintaining effective recommendation performance has become a critical research problem that requires urgent attention.

Current research has explored various approaches to address the cold-start problem, yielding notable progress. Early efforts relied on simple techniques such as dropout [30, 43] to tackle the cold-start issue. Such approaches mitigate over-reliance on historical user data by randomly dropping certain input features, enhancing the generalization ability of models to cold-start users. Some researchers introducing contrastive loss [7, 38], strengthen cold-start user features through the use of positive and negative samples, improving recommendation performance for new users. The application of meta-learning has also become an important research direction for solving the cold-start problem [8, 22]. Such methods enable models to adapt to new, unseen users by leveraging prior knowledge from related tasks or domains, mitigating the impact of sparse historical interaction data [8, 16]. Similarly, approaches based on zero-shot learning utilize transfer learning to adapt knowledge from existing users to new users. Furthermore, generative adversarial networks [40] and graph neural networks [4, 17] have also been explored to tackle the cold-start problem.

However, current methods for the cold-start problem are constrained by the need for specific model structures, limiting their compatibility with advanced sequential recommender models. This



This work is licensed under a Creative Commons Attribution 4.0 International License. *WSDM '26, Boise, ID, USA.*

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2292-9/2026/02

<https://doi.org/10.1145/XXXXXX.XXXXXX>

arises from significant differences in model architecture and feature representation [6, 20], which prevent cold-start solutions from fully leveraging the advantages of advanced models in capturing long-term user behavior patterns and complex preference relationships. These solutions often require retraining from scratch, wasting computational resources, and failing to efficiently utilize existing knowledge and feature representations. Additionally, the limited behavioral data available for cold-start users, often restricted to basic information such as age and gender [9, 18], fails to provide sufficient predictive insights.

Recently, Diffusion Models (DMs) [5, 41] have made significant strides in generative tasks and have emerged as a key research area in deep learning. DMs operate on a probabilistic process where the fundamental concept involves simulating the transition of data from an ordered to a disordered state, and then reversing this process. Consequently, in the realm of recommender systems, DMs can gradually establish and refine user preference profiles based on minimal user data [19, 24, 32, 36]. This process allows the model to predict and reconstruct potential interests and preferences of users from initial sparse information. Therefore, employing DMs to utilize the scarce features of cold-start users to generate input features that align with advanced sequential recommender models holds great potential.

Based on the discussion above, we propose CDMRec, a Causal Conditional Diffusion Model for user cold-start sequential RECommendation. CDMRec generates diffusion behavioral preferences for cold-start users, simulating missing historical interactions to replace their sequential recommendation inputs, enhancing cold-start performance. CDMRec first extracts long-term preference vectors from the historical interactions of warm users to preserve the core interests of users while effectively filtering out noise and irrelevant click data, providing high-quality samples for the diffusion process. Unlike traditional diffusion models, which generate inconsistent behaviors, CDMRec introduces causal inference to infer latent variables that causally influence user preferences. To ensure these variables are both independent and interpretable, we apply mutual information regularization and verify their causal effectiveness through the constructed Preference-Dominant Sequence (PDS). Subsequently, we employ cross-attention in each denoising step to guide the generation of diffusion behavior preferences for cold-start users based on causal variables. CDMRec retains the strengths of advanced sequential recommender models while overcoming cold-start limitations. Additionally, since the model learns extensive user behavior patterns during the diffusion process, it effectively resolves the challenges of limited data and the difficulty in accurately modeling cold-start users.

The main contributions of our study are as follows:

- We propose CDMRec, a Causal Conditional Diffusion Model for user cold-start sequential recommendation. By learning user behavior patterns and generating diffusion behavioral preferences for cold-start users, CDMRec is theoretically compatible with any sequential recommender model, enhancing the cold-start performance.
- By constructing PDS, we mitigate the impact of interference and noise on the diffusion generation process. PDS extracts the interactions that most accurately capture user

authentic preferences, ensuring that the diffusion behavioral preferences reflect the long-term interests and authentic inclinations of users.

- We conduct comprehensive experiments on three public datasets, demonstrating that CDMRec outperforms SOTA methods in cold-start recommendation scenarios. Furthermore, the Compatibility Study highlights the strong compatibility of CDMRec.

2 Preliminary

We provide a brief overview of the problem definitions and notations used in this paper, aiming to assist readers in grasping the core concepts and methods presented later. First, we review the basic concepts of traditional sequential recommendation and the objectives of the models. Then, we discuss the fundamental concepts of diffusion models (DMs) and their theoretical foundations.

2.1 Sequential Recommendation

The primary goal of a sequential recommender system is to speculate the next item a user is likely to engage with, based on their historical interaction sequence. The sequential recommender model learns the behavior patterns of users, extracting latent preferences from the behavior sequence, and predicts the probability of the user engaging with an item from the item set.

Given a user set U and an item set I , for each user $u \in U$, the historical interaction sequence can be defined as $S^u = \{i_1^u, i_2^u, \dots, i_{n^u-1}^u\}$, where $i_j^u \in I$, $1 \leq j \leq n^u - 1$, and n^u denotes the count of items interacted with by user u . The last item interacted with by the user $i_{n^u}^u$ is considered the target for the recommendation task, meaning the model needs to forecast the items that user u is likely to be interested in. Consequently, the objective of a sequential recommendation can be expressed as:

$$\arg \max P(y^u = i_j | S^u), \quad (1)$$

where y^u represents the next interaction item for user u . Specifically, the recommender model calculates the probability of the user interacting with each item in the item set I , based on the behavior sequence S^u , and selects the item with the highest probability, or the top few items, as the recommendation result.

Clearly, this recommendation method heavily relies on the user's historical interaction sequence S^u , and therefore faces a significant cold-start problem when dealing with cold-start users, leading to a considerable decrease in recommendation quality.

In this paper, to better leverage the DM to generate user behavior preferences, the model encodes each item in the behavior sequence S^u individually, rather than encoding the entire sequence S^u as a single vector.

2.2 Diffusion Model

DMs [41] have achieved remarkable success across various fields. They work by converting data into noise and gradually reconstructing the original samples through a reverse process. To facilitate a clearer understanding of the approach introduced in this paper for the readers, we will first provide a brief introduction to traditional DMs. DMs primarily consist of two key processes: the Forward Process and the Reverse Process.

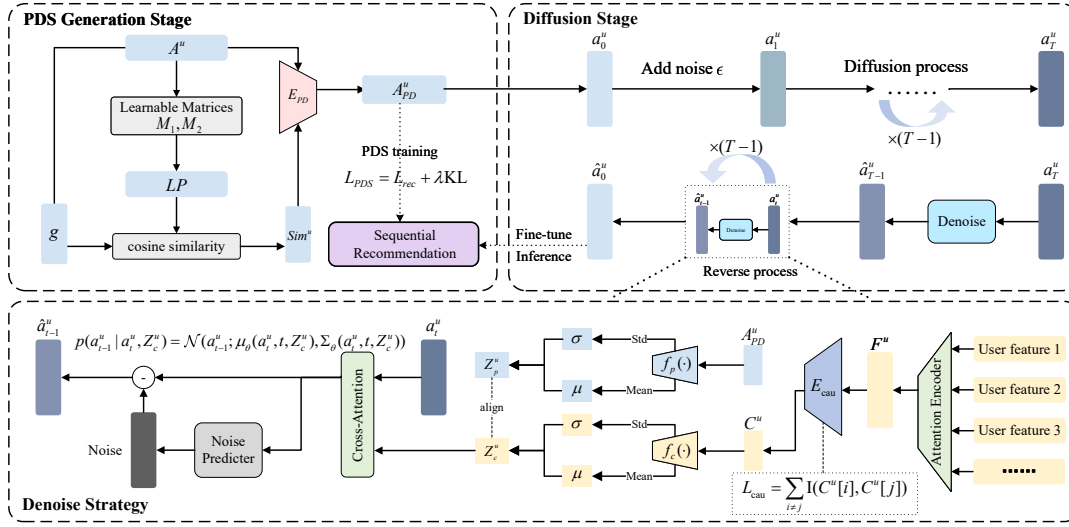


Figure 1: The overall architecture of CDMRec.

Forward Process. The forward process of a DM is a Markov process in which noise is incrementally introduced to the data, making it progressively closer to standard Gaussian noise. Let the data sample be $x_0 \sim q(x_0)$, which is transformed into noise x_T through a diffusion process over T time steps. At each time step, the original data is gradually corrupted with noise, and as $T \rightarrow \infty$, x_T follows a Gaussian distribution. Due to the lengthy noise addition process and the complexity of the expression for β_t , we use the reparameterization technique [13]. Based on the additivity of the Gaussian distribution, we can compute x_t from x_0 in one step:

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\alpha_t}x_0, (1 - \alpha_t)I), \quad (2)$$

where $t \in \{1, 2, \dots, T\}$ represents the number of noise additions applied, $\beta_t \in (0, 1)$ is a hyperparameter controlling the noise intensity, $\mathcal{N}$ denotes the Gaussian distribution, and I representing the covariance matrix of the noise.

Reverse Process. The goal of this process is to learn how to gradually reconstruct the original sample from Gaussian noise. The model starts from the noise x_T and gradually recovers the original data x_0 through a reverse process. The objective is to gradually denoise until the original sample x_0 is recovered. The entire reverse process can be expressed as:

$$p_\theta(x_0, x_1, \dots, x_{T-1}|x_T) = \prod_{t=1}^T p_\theta(x_{t-1}|x_t), \quad (3)$$

where θ represents the neural network parameters, $\mu_\theta(x_t, t)$ and $\Sigma_\theta(t)$ represent the mean and variance predicted by the neural network, respectively.

Optimization. The optimization objective of DM is to generate data that is as similar as possible to x_0 . Typically, DM is optimized by maximizing the Evidence Lower Bound (ELBO). The objective of ELBO is to ensure that the data distribution generated by the model closely approximates the real data distribution. Specifically, our optimization goal for generation is to minimize the following

function:

$$\mathbb{E}_{q(x_T|x_0)q(x_0)} \left[\sum_{t=2}^T D_{KL}(q(x_{t-1}|x_t, x_0) || p_\theta(x_{t-1}|x_t)) \right], \quad (4)$$

where D_{KL} represents the Kullback-Leibler divergence, which ensures that the data distribution p_θ generated by the model closely approximates the real data distribution q . According to [13], the objective loss can be simplified and expressed as:

$$L_{simple} = \mathbb{E}_{t, x_0, \epsilon} [||\epsilon - \epsilon_\theta(\sqrt{\alpha_t}x_0 + \sqrt{1 - \alpha_t}\epsilon), t||^2]. \quad (5)$$

3 Method

3.1 Overview

To address the user cold-start problem in sequential recommendation, we propose CDMRec model, which leverages user features to diffuse user interaction preferences from a causal perspective. The overall architecture is shown in Figure 1. Specifically, CDMRec first constructs PDS to capture users' core preferences, then leverages a causality-aware diffusion process guided by user features to generate behavior preference sequences. These generated preferences can be seamlessly integrated into existing sequential recommenders, enabling effective recommendation even for users with limited interaction history.

3.2 Preference-Dominant Sequence Generation

To reduce the influence of irrelevant interactions (such as mislicked items) on diffusion generation, we construct a PDS by identifying historical items that reflect a user's true long-term preferences.

Given a historical interaction sequence S^u of user u , we embed the sequence into an embedding matrix $A^u \in \mathbb{R}^{(n^u-1) \times d}$, where d is the dimension of the item embedding. The encoding of the j -th item in the interaction sequence i^u_j is represented by $A^u[j]$. The embedding of the real recommendation value $i^u_{n^u}$ is denoted as $\mathcal{G} \in \mathbb{R}^{1 \times d}$.

To infer the user's overall preference, we construct learnable matrices $M_1 \in \mathbb{R}^{\frac{d}{2} \times d}$ and $M_2 \in \mathbb{R}^{1 \times \frac{d}{2}}$, and use the following formula to transform the matrix A^u :

$$LP = \text{softmax}(M_2 \tanh(M_1 A^{uT})) A^{uT}, \quad (6)$$

where $LP \in \mathbb{R}^{1 \times d}$. Similar to Attention mechanisms, this formula aggregates the features of all items in the sequence into a single vector, which represents the long-term preferences of the user u . Then, we compute the similarity between each item in S^u and LP . Similarly, we also consider the similarity between each item and the real recommendation value. Considering both types of similarity, we assign a score to each item. The calculation formula of Sim_i^u can be expressed as:

$$Sim_i^u = \mathcal{S} \cos_{\text{sim}}(i_j, LP) + (1 - \mathcal{S}) \cos_{\text{sim}}(i_j, \mathcal{G}), \quad (7)$$

where $1 \leq j \leq n^u - 1$, $\mathcal{S} \in (0, 1)$ is a hyperparameter used to adjust the weight between the two types of similarity.

Given historical interaction embedding A^u , and the similarity values $Sim^u = \{Sim_1^u, Sim_2^u, \dots, Sim_{n^u-1}^u\}$ computed based on similarity, we treat Sim^u as a new dimension of A^u , resulting in A_{sim}^u as the input to the encoder:

$$A_{sim}^u = [A^u, Sim^u] \in \mathbb{R}^{(n^u-1) \times (d+1)}. \quad (8)$$

After obtaining the fused feature matrix A_{sim}^u , we process it through the encoder E_{PD} to extract the PDS $A_{PD}^u = E_{PD}(A_{sim}^u)$, $A_{PD}^u \in \mathbb{R}^{l \times d}$, where l is a hyperparameter used to control the length of the PDS. This result represents the portion of historical interaction sequence A^u that most accurately reflects the long-term preferences of the user u .

Generally, any existing sequential recommender models can be integrated with CDMRec. Suppose the model predicts the interaction probability for user u with item $i_{n^u}^u$ as $g(i_{n^u}^u)$, and the ground truth is $y_{n^u}^u \in \{0, 1\}$ (0 for no interaction, 1 for interaction). The loss can be expressed as:

$$L_{\text{rec}} = -\frac{1}{N} \sum_{u \in U_{\text{cur}}} (y_{n^u}^u \log(g(i_{n^u}^u)) + (1 - y_{n^u}^u) \log(1 - g(i_{n^u}^u))), \quad (9)$$

where N is the batch size, and U_{cur} is the set of all users in current batch.

3.3 Diffusion Process

Forward process. In this process, we add noise to the data samples. In this study, we use PDS A_{PD}^u , as the initial data sample a_0^u for user u , and perform diffusion generation based on it. Given the PDS A_{PD}^u of user u , we set it as the initial state $a_0^u = A_{PD}^u$. Then, based on the additivity of the Gaussian distribution, we add multiple segments of Gaussian noise to a_0 at once:

$$q(a_t^u | a_0^u) = \mathcal{N}(a_t^u; \sqrt{\alpha_t} a_0^u, (1 - \alpha_t) I), \quad (10)$$

where a_t^u represents the data at time step t , $\alpha_t = \prod_{j=1}^t \alpha_j$, $\alpha_j = 1 - \beta_j$, where $\beta_j \in (0, 1)$ is a hyperparameter that controls the intensity of the noise. I is the identity matrix, representing the covariance matrix of the noise. In this process, the control of noise intensity is achieved by adjusting the hyperparameter β_j . Specifically, following previous works [13, 26], we set β_j as a linearly

increasing constant from 10^{-4} to 0.05. This strategy guarantees that during the noise adding process, as t increases, the noise intensity gradually intensifies, progressively transforming the data into Gaussian noise.

Causality-aware reverse process. In this process, we reconstruct the original data a_0^u from the high-noise sample by gradual denoising. To make the recovery more accurate and consistent with user features, we are inspired by Rombach et al. [26] and incorporate cross-attention to guide this process. To improve the semantic fidelity and interpretability of the diffusion behavior preferences generated in cold-start scenarios, we propose a novel causality-aware reverse process that integrates causal representation learning into the denoising stage.

First, we use an Attention encoder to encode the features of user u , obtaining the embedding F^u . To extract the core causal variables from F_u , we design an encoder E_{cau} , which learns a disentangled causal representation. $C^u = E_{\text{cau}}(F_u)$ is the learned latent vector, representing the underlying causal factors of user behavior.

To facilitate disentanglement and avoid redundancy between C^u dimensions, i.e., multiple dimensions preserve the same causal features, we introduce a sparse mutual information regularizer:

$$L_{\text{cau}} = \sum_{i \neq j} \text{I}(C^u[i], C^u[j]), \quad (11)$$

where $\text{I}(\cdot, \cdot)$ denotes mutual information. Minimizing this term encourages each dimension of C^u to capture distinct causal effects. To approximate $\text{I}(C^u[i], C^u[j])$, we adopt the Mutual Information Neural Estimator [2]. Specifically, we introduce a neural discriminator T_ϕ and calculate the estimated I as:

$$\begin{aligned} \text{I}(C^u[i], C^u[j]) &\approx \mathbb{E}_{p_{ij}}[T_\phi(x, y)] \\ &\quad - \log \mathbb{E}_{p_{i \otimes j}}[\exp(T_\phi(x', y'))], \end{aligned} \quad (12)$$

where $(x, y) \sim p_{ij}(C^u[i], C^u[j])$ represents the joint distribution sample, $p(C^u[i], C^u[j])$, and $(x', y') \sim p_i(C^u[i])p_j(C^u[j])$ is sampled from the marginal distributions of the two variables respectively.

However, disentanglement alone does not guarantee that C^u captures the true causal information. Therefore, we use A_{PD}^u to verify the causal validity of C^u . Since their semantic distributions are different, forcing alignment in the original vector space may be problematic, we compare them in a shared latent space learned by an aligned encoder. $f_c(\cdot)$ and $f_p(\cdot)$ are two multi-layer projection networks that map C^u and A_{PD}^u to latent distributions:

$$\begin{aligned} Z_c^u &\sim \mathcal{N}(\mu_c(C^u), \Sigma_c(C^u)), \\ Z_p^u &\sim \mathcal{N}(\mu_p(A_{PD}^u), \Sigma_p(A_{PD}^u)). \end{aligned} \quad (13)$$

To ensure meaningful alignment, we define the loss as the distance between these two Gaussian distributions:

$$\begin{aligned} L_{\text{align}} &= \mathbb{E}_u [\text{KL}(\mathcal{N}(\mu_c, \Sigma_c) \parallel \mathcal{N}(\mu_p, \Sigma_p)) \\ &\quad + \text{KL}(\mathcal{N}(\mu_p, \Sigma_p) \parallel \mathcal{N}(\mu_c, \Sigma_c))], \end{aligned} \quad (14)$$

where $\text{KL}(\cdot \parallel \cdot)$ denotes the KL divergence between two multivariate Gaussians. This distribution-level alignment encourages the semantic information contained in the causal variables to be consistent with the behavioral distribution implied by the PDS.

Given a noisy sample a_t^u at time step t , our goal is to infer a_t^u at the previous time step $t-1$, i.e., a_{t-1}^u . Unlike traditional DMs, we introduce Z_c^u to guide the sample prediction. The reverse process is probabilistically expressed as:

$$p(a_{t-1}^u | a_t^u, Z_c^u) = \mathcal{N}(a_{t-1}^u; \mu_\theta(a_t^u, t, Z_c^u), \Sigma_\theta(a_t^u, t, Z_c^u)), \quad (15)$$

where $\mu_\theta(a_t^u, t, Z_c^u)$ and $\Sigma_\theta(a_t^u, t, Z_c^u)$ are the mean and covariance predicted by a neural network parameterized by θ , based on a_t^u and Z_c^u .

To incorporate the causal variable Z_c^u into the reverse process, we apply a cross-attention mechanism at each denoising step t , where the current noisy sample a_t^u serves as the query, and Z_c^u provides the keys and values:

$$\text{Cross-Attention}(a_t^u, Z_c^u) = \text{softmax}\left(\frac{Q_t K^T}{\sqrt{d}}\right) V, \quad (16)$$

where d is the vector dimension, with projections defined as:

$$Q_t = M_Q^{(I)} \phi^{(I)}(a_t^u), \quad K = M_K^{(I)} Z_c^u, \quad V = M_V^{(I)} Z_c^u, \quad (17)$$

where $\phi^{(I)}$ is the noise predictor at step for the current iteration, and $M_Q^{(I)}$, $M_K^{(I)}$, and $M_V^{(I)}$ are learnable matrices.

3.4 Optimization

PDS generation training. We employ the cross-entropy function L_{rec} to optimize the gap between the predicted values and the real labels, which will not be elaborated further here. In addition to the standard cross-entropy loss, to further ensure that the generated PDS does not deviate too much from conventional item sequence encoding, we introduce a regularization term in L_{rec} , which measures the distributional difference between PDS A_{PD}^u and historical interaction sequence A^u . Specifically, we employ the KL divergence as a statistical method to quantify the distance between these two distributions. The loss function for this stage is defined as the weighted sum of the cross-entropy loss, which ensures accurate predictions, and the KL divergence regularization term:

$$L_{PDS} = L_{rec} + \lambda \text{KL}(P(A_{PD}^u) || P(A^u)), \quad (18)$$

where hyperparameter λ controls the weight of the regularization term, $P(A_{PD}^u)$ and $P(A^u)$ represent the distributions of the PDS and the historical interaction sequence. By minimizing L_{PDS} , we are able to not only optimize the recommendation performance but also ensure that the generated PDS accurately captures and reflects the long-term interests and real preferences of users.

Diffusion training. To recover a_0^u , we aim to maximize the marginal likelihood $p(a_0^u)$. However, directly computing this marginal likelihood is infeasible, so we use variational inference to approximate it. The objective of variational inference is to approximate the real posterior distribution $p(a_1^u, \dots, a_T^u | a_0^u, Z_c^u)$ by introducing a variational distribution $q(a_1^u, \dots, a_T^u | a_0^u, Z_c^u)$, and training by optimizing ELBO which can be expressed as:

$$\log p(a_0^u) \geq \mathbb{E}_{q(a_1^u, \dots, a_T^u | a_0^u, Z_c^u)} \left[\log \frac{p(a_0^u, a_1^u, \dots, a_T^u)}{q(a_1^u, \dots, a_T^u | a_0^u, Z_c^u)} \right]. \quad (19)$$

According to [13], the conditional diffusion loss function can be expressed as follows:

$$L_{CDM} = \mathbb{E}_{t, a_0^u, \epsilon} [\|\epsilon - \epsilon_\theta(\sqrt{\alpha_t} a_0^u + \sqrt{1 - \alpha_t} \epsilon, t, Z_c^u)\|^2], \quad (20)$$

where $\epsilon \in \mathcal{N}(0, 1)$ represents the actual noise added to the sample, which is the target the model aims to reconstruct. $\epsilon_\theta(\sqrt{\alpha_t} a_0^u + \sqrt{1 - \alpha_t} \epsilon, t, Z_c^u)$ is the noise predicted by the model. By minimizing L_{CDM} , the model can accurately remove noise from the noisy samples and recover the original data a_0^u covered by noise, based on the causal variable Z_c^u .

The total loss of the diffusion training becomes:

$$L_{diff} = L_{CDM} + \beta_1 L_{cau} + \beta_2 L_{align}, \quad (21)$$

where β_1 and β_2 control the balance between feature dimension separation, causal variable extraction, and diffusion preference reconstruction. L_{diff} encourages the model to generate diffusion behavior preferences that are both consistent with the data and based on causal relationships.

Inference. For warm users, fewer steps of diffusion are performed based on their behavior to ensure that the sequence information is still preserved in a_T^u . The reverse process is then carried out with the help of Z_c^u , obtaining the user's diffusion behavior preferences $\hat{a}_0^u$, which is used for recommendation. For cold-start users, noise is sampled from Gaussian noise, and the reverse process is conducted with Z_c^u to directly generate their diffusion behavior preferences $\hat{a}_0^u$. This generated sequence replaces the missing behavior sequence and is used for recommendation.

4 Experiment

In this section, we conducted comparative experiments and ablation studies on three public datasets to answer the succeeding three research questions:

- **RQ1:** Whether CDMRec achieves the best recommendation performance in both user cold-start and regular scenarios compared to other baseline methods?
- **RQ2:** Can CDMRec effectively improve the cold-start recommendation performance of sequential recommender models, and does it have any impact on the recommendation performance in regular scenarios?
- **RQ3:** Whether the components of CDMRec work as intended and what impact does the absence of certain components have on overall performance?

4.1 Experimental Setup

Datasets. To evaluate CDMRec, we carried out experiments on three public datasets: KuaiRec [10], XING [1], and Douban [25]. KuaiRec, from the mobile video platform Kuaishou, includes 1,141 users, 3,327 items, and 4,676,570 interactions. XING, extracted from the ACM RecSys 2017 competition, contains 106,881 users, 20,519 job positions, and 4,306,183 interactions, with job positions represented by a 2,738-dimensional vector. Douban, from the Chinese social platform, provides data from 2,732 users and 16,967 items after excluding anomalous users.

For all datasets, we randomly selected 30% of the users as the test set, with half of them untouched and the other half having their historical interaction sequences removed to simulate cold-start users. The remaining 70% of the users were split into training and validation sets in an 8:2 ratio. For each interaction sequence,

Table 1: Recommendation performance comparison between CDMRec and baselines. The improvements are calculated by comparing CDMRec with the ranked second (underlined). The performance improvement of CDMRec in recommendation is statistically significant, with a p-value less than 0.01 according to the t-test.

Method	Regular Recommendation						Cold-Start Recommendation					
	KuaiRec		Douban		XING		KuaiRec		Douban		XING	
	Recall	NDCG	Recall	NDCG	Recall	NDCG	Recall	NDCG	Recall	NDCG	Recall	NDCG
DeepMusic	0.0412	0.0283	0.0039	0.0083	<u>0.4102</u>	0.2949	0.0396	0.0193	0.0026	0.0057	0.2687	0.1601
DropoutNet	0.0369	0.0273	0.0042	0.0102	0.3032	0.2184	0.0309	0.0140	0.0031	0.0063	0.2770	0.1958
MeLU	0.0390	0.0292	0.0041	<u>0.0119</u>	0.3032	0.3245	0.0415	0.0161	0.0035	0.0061	0.2839	0.2300
PDRec	0.0356	0.0282	0.0042	0.0096	0.3161	0.2518	0.0297	0.0075	0.0013	0.0026	0.0985	0.1355
SDCRec	<u>0.0416</u>	<u>0.0350</u>	0.0037	0.0098	0.3420	0.2947	<u>0.0492</u>	0.0187	0.0024	<u>0.0071</u>	0.3015	0.2032
PDMA	0.0326	0.0318	0.0039	0.0084	0.3821	0.2884	0.0411	0.0211	0.0033	0.0065	<u>0.3676</u>	0.2419
MACDR	0.0398	0.0289	<u>0.0045</u>	0.0103	0.3053	<u>0.3420</u>	0.0423	<u>0.0223</u>	<u>0.0037</u>	0.0064	0.2908	<u>0.2494</u>
C2lRec	0.0402	0.0307	0.0036	0.0105	0.3025	0.3381	0.0425	0.0217	0.0035	0.0068	0.3514	0.2426
CDMRec	0.0423	0.0353	0.0047	0.0124	0.4183	0.3512	0.0564	0.0251	0.0039	0.0076	0.3823	0.2561
improv.	1.68%	0.86%	4.44%	4.20%	1.97%	2.69%	14.63%	12.56%	5.41%	7.04%	4.00%	2.69%

assuming the sequence length is n , we used the first $n - 1$ interactions as the model input, with the last interaction serving as the recommendation ground truth for training.

Evaluation Metrics. We conducted comparative experiments between CDMRec and baseline models in both cold-start recommendation scenarios (with cold-start users) and regular scenarios (with warm users). The evaluation was based on two widely used metrics: Recall@ K and Normalized Discounted Cumulative Gain (NDCG)@ K . Recall@ K indicates the proportion of relevant items the model is able to identify, while NDCG@ K takes into account the ranking of the recommended items, giving higher weight to relevant items ranked higher. In this study, following previous works [14, 38], we set K to 20 and reported the average result across all users in U . Each experiment was run independently five times, and the final results were averaged.

Baselines. To evaluate the effectiveness of CDMRec for recommendation to both cold-start and warm users, we compared it with three datasets and seven baseline models.

- **DeepMusic** [29] addresses the cold-start problem by predicting latent factors from music audio and achieves superior recommendation performance over traditional methods employing deep convolutional neural networks.
- **DropoutNet** [30] is a neural network-based latent model that addresses the cold-start problem in recommender systems through dropout optimization and demonstrates SOTA accuracy on several public datasets.
- **MeLU** [16] estimates user preferences by leveraging a small number of consumed items and provides an evidence-based candidate selection strategy to tackle the cold-start problem.
- **PDRec** [24] is a plug-in diffusion recommendation framework that effectively alleviates data sparsity and improves recommendation performance through full-item preference modeling, positive and negative sample enhancement, and historical behavior reweighting.

- **SDCRec** [7] tackles the cold-start problem and reduces popularity bias by utilizing dynamic node embeddings and a dual-branch self-supervised contrastive objective, fully exploiting social relationships and item similarities.
- **PDMA** [31] addresses cold-start users' preference learning by dividing it into common preference transfer and novel preference adaptation, mitigating memory overfitting through the injection of attribute noise.
- **MACDR** [33] introduces the MACDR model, which solves cross-domain cold-start recommendation issues by enhancing the Mixture-Of-Experts mapping function with prototypes and optimizing preference distribution alignment, addressing user behavior pattern similarity and sparse supervision signal problems.
- **C2lRec** [39] is a causal contrastive learning recommendation framework that achieves accurate recommendations for cold-start users by mining primary dependent variables from social circles and modeling hidden preferences.

Implementation Details. We implemented all baseline models using the source code provided in the original papers. In the comparative experiments, CDMRec uses SASRec [15] as the underlying sequential model. The item embedding dimension is set to 64, and the Adam optimizer is employed with a learning rate set to 0.0001. The hyperparameters are set as follows: S is 0.7, l is 6, λ is 0.5, β_1 is 1, and β_2 is 1. To be fair, we trained all baseline models using the default hyperparameters. All experiments were conducted on a server equipped with an NVIDIA L40S GPU, an Intel Core i9-14900K CPU, 128 GB of RAM, running Ubuntu 20.04.

4.2 Performance Comparison (RQ1)

As shown in Table 1, DMCGRec was comprehensively compared with seven baseline methods in terms of metrics across three datasets (KuaiRec, Douban, XING) under both regular recommendation and cold-start recommendation scenarios. The baseline methods include a generative model (DeepMusic), a Dropout model (DropoutNet), two meta-learning models (MeLU, PDMA), a contrastive learning

Table 2: Performance comparison of SASRec, CL4SRec and SMLP4Rec with and without CDMRec. CDMRec significantly enhances recommendation performance in cold-start scenarios without compromising the regular recommendation performance of the original sequential recommendation model.

Method	Regular Recommendation						Cold-Start Recommendation					
	KuaiRec		Douban		XING		KuaiRec		Douban		XING	
	Recall	NDCG	Recall	NDCG	Recall	NDCG	Recall	NDCG	Recall	NDCG	Recall	NDCG
SASRec	0.0411	0.0337	0.0045	0.0118	0.4152	0.3406	0.0409	0.0219	0.0033	0.0067	0.3301	0.2205
+CDMRec	0.0423	0.0353	0.0047	0.0124	0.4183	0.3512	0.0564	0.0251	0.0039	0.0076	0.3823	0.2561
improv.	2.92%	4.75%	4.44%	5.08%	0.75%	3.11%	37.90%	14.61%	18.18%	13.43%	15.81%	16.15%
CL4SRec	0.0412	0.0338	0.0043	0.0115	0.4008	0.3303	0.0406	0.0212	0.0038	0.0059	0.2507	0.1804
+CDMRec	0.0416	0.0342	0.0045	0.0129	0.4053	0.3355	0.0502	0.0244	0.0042	0.0068	0.3204	0.2009
improv.	0.97%	1.18%	4.65%	12.17%	1.12%	1.57%	23.65%	15.09%	10.53%	15.25%	27.80%	11.36%
SMLP4Rec	0.0419	0.0342	0.0042	0.0115	0.3954	0.3394	0.0325	0.0276	0.0028	0.0051	0.2703	0.2102
+CDMRec	0.0424	0.0351	0.0046	0.0119	0.4164	0.3492	0.0416	0.0312	0.0037	0.0063	0.3703	0.2407
improv.	1.19%	2.63%	9.52%	3.48%	5.31%	2.89%	28.00%	13.04%	32.14%	23.53%	37.00%	14.51%

model (SDCRec), a zero-shot learning model (LLAE), and an expert mixture model (MACDR). In Table 1, the highest accuracy among the seven baseline methods is underlined, and the performance improvement of DMCGRec is determined by comparing it with these top-performing baselines. Overall, DMCGRec demonstrates outstanding performance in most cases, particularly exhibiting significant advantages in the cold-start scenario.

In regular recommendation settings, DMCGRec achieves comparable or superior Recall and NDCG performance on KuaiRec and Douban, with a notable 4.2% NDCG gain on Douban, demonstrating its stronger ability to capture users' long-term preferences. This improvement stems from the PDS mechanism, which filters informative interactions and suppresses noise, thereby refining user modeling.

In the cold-start scenario, DMCGRec shows even greater advantages, improving Recall by 14.63%, 5.41%, and 4.00% on KuaiRec, Douban, and XING, accompanied by NDCG gains of 12.56%, 7.04%, and 2.69%, respectively. These improvements arise from its diffusion-based generation of preference sequences for unseen users, effectively compensating for the absence of historical interactions.

In summary, DMCGRec demonstrates exceptional performance in both two scenarios through. In cold-start scenarios, DMCGRec significantly enhances the accuracy and robustness of the recommender model by generating high-quality behavior preference sequences. These results fully validate the effectiveness of DMCGRec in addressing the user cold-start problem and improving recommendation performance, comprehensively answering the **RQ1**.

4.3 Compatibility Study (RQ2)

To evaluate the compatibility of CDMRec with various sequential recommender models and its impact on recommendation performance, we conducted experiments on three sequential recommender models: SASRec [15], CL4SRec [37], and SMLP4Rec [11]. These experiments were carried out under both regular recommendation and cold-start recommendation scenarios. Table 2 presents the Recall and NDCG metrics for each model across different datasets

(KuaiRec, Douban, XING), as well as the performance improvements achieved by integrating CDMRec.

In the regular recommendation setting, CDMRec consistently improves the performance of various sequential models. For SASRec, Recall and NDCG increase by 2.92% and 4.75%, showing that CDMRec strengthens its ability to model long-term preferences. CL4SRec also benefits from CDMRec, with gains of 0.97% in Recall and 1.18% in NDCG, while SMLP4Rec achieves improvements of 1.19% and 2.63%, respectively. These results confirm that CDMRec provides stable enhancement across different model architectures.

In the cold-start scenario, CDMRec demonstrates even more substantial effectiveness. SASRec achieves boosts of 37.90% in Recall and 14.61% in NDCG, while CL4SRec improves by 23.65% and 15.09%. Similarly, SMLP4Rec achieves 28.00% improvement in Recall and 13.04% improvement in NDCG. These significant improvements indicate that CDMRec generates high-quality diffusion preference sequences from limited user attributes, effectively compensating for missing interaction histories and significantly enhancing the accuracy of cold-start recommendations.

These results demonstrate that CDMRec can be flexibly integrated into different sequential recommendation frameworks. By optimizing user preference modeling and implementing effective noise handling mechanisms, CDMRec significantly boosts the overall performance of recommender systems. Therefore, while CDMRec enhances the cold-start recommendation performance of sequential recommender models, it also positively impacts regular recommendation scenarios. These findings comprehensively address **RQ2**, validating CDMRec's effectiveness and versatility across diverse application contexts.

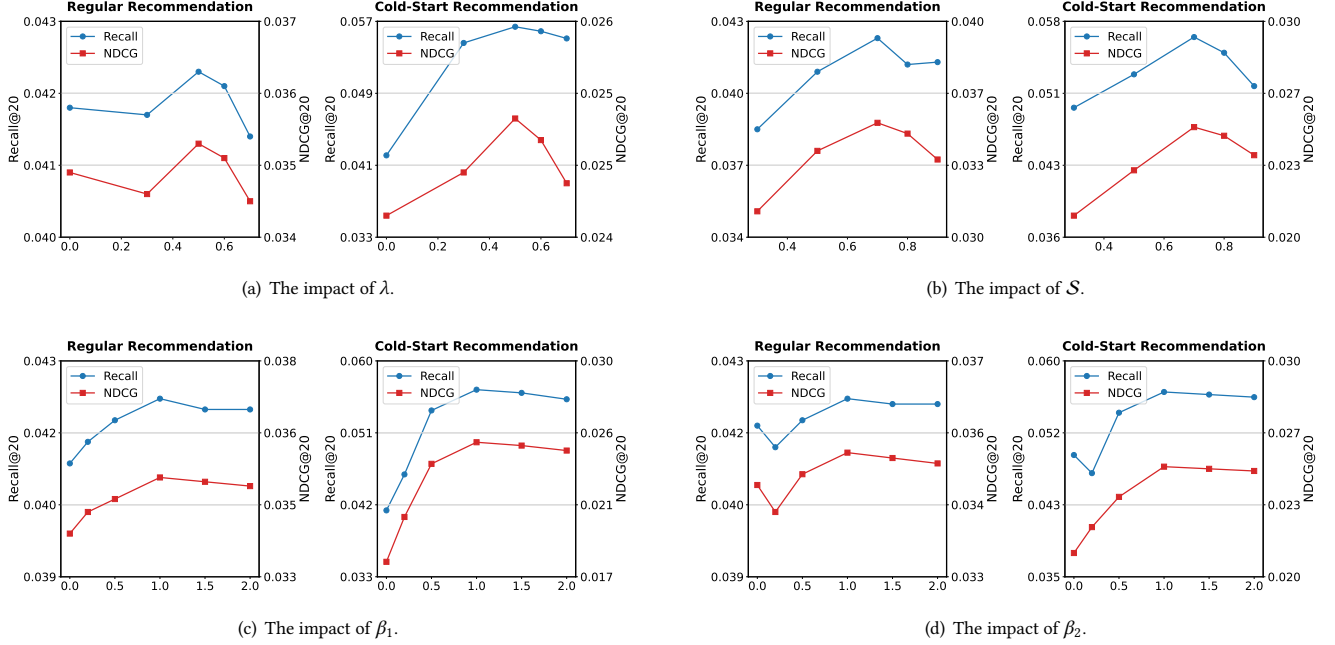
4.4 Ablation Study (RQ3)

To investigate the impact and effectiveness of each component of CDMRec on the overall Recommendation performance, we designed three variant models:

- **CDMRec-w/o Sim:** During the PDS generation phase, item similarities are not calculated. The last few items in the user interaction sequence are directly selected as the PDS.

Table 3: Ablation study results of CDMRec and the variants.

Method	Regular Recommendation						Cold-Start Recommendation					
	KuaiRec		Douban		XING		KuaiRec		Douban		XING	
	Recall	NDCG	Recall	NDCG	Recall	NDCG	Recall	NDCG	Recall	NDCG	Recall	NDCG
CDMRec-w/o Sim	0.0403	0.0345	0.0036	0.0111	0.4033	0.3417	0.0394	0.0229	0.0033	0.0064	0.3620	0.2542
CDMRec-w/o CA	0.0421	0.0327	0.0040	0.0122	0.4126	0.3490	0.0264	0.0205	0.0025	0.0035	0.2158	0.1953
CDMRec-w/o KL	0.0418	0.0349	0.0038	0.0106	0.3975	0.3369	0.0421	0.0242	0.0036	0.0062	0.3571	0.2444
CDMRec	0.0423	0.0353	0.0047	0.0124	0.4183	0.3512	0.0564	0.0251	0.0039	0.0076	0.3823	0.2561

**Figure 2: The impact of varying the hyperparameters on experimental results on the dataset KuaiRec.**

- **CDMRec-w/o CA:** The cross-attention mechanism in the diffusion reverse process is removed, preventing user features from influencing the generation of the diffusion behavior preferences.
- **CDMRec-w/o KL:** The regularization hyperparameter λ is set to 0, eliminating the KL divergence constraint that aligns the PDS with the distribution of the user’s historical interaction sequences.

Table 3 presents the Recall and NDCG metrics for the three variant models alongside the complete CDMRec model across KuaiRec, Douban, and XING under both regular and cold-start recommendation scenarios. It is evident that all variants exhibit lower performance than the complete model in both settings, demonstrating that each module plays a crucial role in enhancing the effectiveness of CDMRec. These results further indicate that removing any component weakens preference extraction and ultimately reduces overall recommendation quality. Specifically, **CDMRec-w/o Sim:** By not

calculating item similarities, the model struggles to accurately select interaction items that truly represent the user’s preferences. This limitation weakens the ability to model long-term user preferences, resulting in decreased recommendation performance across all scenarios; **CDMRec-w/o CA:** Removing the cross-attention mechanism prevents user features from guiding the generation of behavior preference sequences. This is particularly detrimental in the cold-start scenario, where it reduces the recommendation accuracy for new users; **CDMRec-w/o KL:** Eliminating KL regularization causes the generated preference sequences to diverge from the distribution of the user’s historical behaviors. This inconsistency affects the model’s stability and generalization capabilities. In summary, the ablation study validates the essential roles of similarity calculation, the cross-attention mechanism, and KL regularization within CDMRec. These components ensure the model’s efficiency and robustness across different recommendation scenarios, further confirming the rationality and effectiveness of CDMRec’s design, comprehensively answering the **RQ3**.

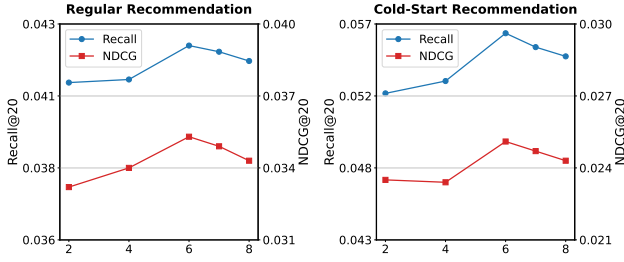


Figure 3: The impact of varying the hyperparameter l on experimental results on the dataset KuaiRec

4.5 Hyperparameters Analysis

To analyze the impact of hyperparameters on CDMRec, we conducted experiments on KuaiRec, which offers a larger user base, higher interaction density, and richer behavior patterns than Douban and XING, making it suitable for sensitivity evaluation. For the sequence length l (Fig.3), both Recall@20 and NDCG@20 for regular and cold-start users increase initially but decline when $l > 6$, as longer sequences hinder the diffusion process. For the regularization weight λ (Fig.2(a)), performance peaks at $\lambda = 0.5$, where moderate distribution alignment improves generalization. Larger values slightly degrade results. For $\mathcal{S}$ (Fig.2(b)), which balances long-term preference and the ground-truth item in PDS construction, performance peaks at $\mathcal{S} = 0.7$. Excessive emphasis on the ground-truth item biases the model toward short-term preferences. For β_1 (Fig.2(c)), governing mutual information regularization, performance peaks at $\beta_1 = 1$. Smaller values fail to capture meaningful causal factors, while larger ones overemphasize disentanglement. For β_2 , controlling alignment between causal variables and PDS behavior, performance improves up to $\beta_2 = 1$ and plateaus thereafter, indicating that moderate alignment ensures semantic consistency without overfitting.

Overall, these results highlight the necessity of balanced hyperparameter settings, where moderate values consistently yield the best trade-off between regular and cold-start performance, while extreme values tend to introduce overfitting or bias.

5 Related Work

5.1 User Cold-start Recommendation

The user cold-start problem is one of the critical challenges that recommender systems urgently need to address [12]. It primarily arises from the difficulty of making effective recommendations for new users who lack sufficient interaction data. To tackle this issue, the academic community has proposed various approaches. Early methods such as DropoutNet [30] utilize neural network-based models combined with dropout to train models that maintain robust performance even when input data is missing. Subsequently, graph neural networks began to be applied in the recommendation domain. For instance, HGNN [21] employs heterogeneous graph convolutional networks and predicts social and semantic links to learn embedding representations of users and items. IHGNN [4] introduces inductive heterogeneous graph neural networks and

incorporates multi-level attention mechanisms to address the sparsity of user attributes. Meanwhile, LLA [18] treats the cold-start problem as a zero-shot learning task and utilizes a low-rank linear autoencoder to mitigate the effects of domain shift and sparse data. Approaches based on meta-learning, such as MeLU [16], adapt quickly to new user preferences with minimal interaction data. Building upon this, FANP [20] adopts probabilistic meta-learning strategies and stream-based adaptive neural processes to enhance the ability to capture dynamic changes in user preferences.

5.2 Diffusion Model Based Recommendation

Recently, Diffusion Models (DMs) have gained popularity in recommender systems due to their generative capabilities and advantages in modeling complex relationships. DiffNet [35], an early work in this field, integrates DMs into recommender systems by optimizing user embeddings through simulating influence propagation in social networks. DiffNet++ [34] incorporates the user-item interest network and employs a multi-level attention mechanism to enhance recommendation performance based on DiffNet. In the realm of sequential recommender systems, DiffuRec [19] represents items as distributions and captures diverse user interests by injecting uncertainty through a diffusion process. DreamRec [42] treats sequential recommendation as a generative task, using guided DMs to generate "oracle" items that reflect user authentic preferences, bypassing the limitations of traditional negative sampling. Diff4Rec [36] improves performance with a curriculum scheduling diffusion augmentation framework for data augmentation and model optimization. PDR [24] employs a plug-in DM to reweight historical behaviors, enhancing both positive and negative samples to address data sparsity and ensure effective model optimization.

Although cold-start methods have made progress, they still face two key limitations: poor compatibility with advanced sequential recommendation models and insufficient ability to capture the authentic preferences of new users. To address these issues, CDMRec leverages diffusion model to ensure both high model compatibility and accurate preference generation for cold-start users.

6 Conclusion

In this paper, we propose CDMRec to address the user cold-start problem in sequential recommendation. By constructing a Preference-Dominant Sequence, CDMRec effectively filters irrelevant user interactions. Leveraging causal inference, CDMRec extracts disentangled meaningful latent variables, which are then used to guide the diffusion process via cross-attention, enabling the generation of behavior preference sequences that align well with user features. Since the diffusion behavior preferences generated by CDMRec closely match the distribution of user behavior sequences, they can replace traditional behavior sequences as input for regular sequential recommender models.

Acknowledgments

This work was supported in part by the National Natural Science Foundation of China under Grant 62372242, Jiangsu Provincial Major Project on Basic Research of Cutting-edge and Leading Technologies, under grant no. BK20232032, and Jiangsu Province Graduate Research Innovation Program under Grant KYCX25_1679.

References

- [1] Fabian Abel, Yashar Deldjoo, Mehdi Elahi, and Daniel Kohlsdorf. 2017. Recsys challenge 2017: Offline and online evaluation. In *Proceedings of the eleventh acm conference on recommender systems (RecSys)*. 372–373.
- [2] Mohamed Ishmael Belghazi, Aristide Baratin, Sai Rajeswar, Sherjil Ozair, Yoshua Bengio, Aaron Courville, and R Devon Hjelm. 2018. Mine: mutual information neural estimation. *arXiv preprint arXiv:1801.04062* (2018).
- [3] Tesfaye Fenta Boka, Zhendong Niu, and Rama Bastola Neupane. 2024. A survey of sequential recommendation systems: Techniques, evaluation, and future directions. *Information Systems* (2024), 102427.
- [4] Desheng Cai, Shengsheng Qian, Quan Fang, Jun Hu, and Changsheng Xu. 2023. User cold-start recommendation via inductive heterogeneous graph neural network. *ACM Transactions on Information Systems* 41, 3 (2023), 1–27.
- [5] Florinel-Alin Croitoru, Vlad Hondru, Radu Tudor Ionescu, and Mubarak Shah. 2023. Diffusion models in vision: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45, 9 (2023), 10850–10869.
- [6] Manqing Dong, Feng Yuan, Lina Yao, Xiwei Xu, and Liming Zhu. 2020. Mamo: Memory-augmented meta-optimization for cold-start recommendation. In *Proceedings of the 26th ACM SIGKDD international conference on knowledge discovery & data mining (SIGKDD)*. 688–697.
- [7] Jing Du, Zesheng Ye, Lina Yao, Bin Guo, and Zhiwen Yu. 2022. Socially-aware dual contrastive learning for cold-start recommendation. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*. 1927–1932.
- [8] Yuntao Du, Xinjun Zhu, Lu Chen, Ziquan Fang, and Yunjun Gao. 2022. Metakg: Meta-learning on knowledge graph for cold-start recommendation. *IEEE Transactions on Knowledge and Data Engineering* 35, 10 (2022), 9850–9863.
- [9] Philip J Feng, Pingjun Pan, Tingting Zhou, Hongxiang Chen, and Chuanjiang Luo. 2021. Zero shot on the cold-start problem: Model-agnostic interest learning for recommender systems. In *Proceedings of the 30th ACM international conference on information & knowledge management (CIKM)*. 474–483.
- [10] Chongming Gao, Shijun Li, Wenqiang Lei, Jiawei Chen, Biao Li, Peng Jiang, Xiangnan He, Jiaxin Mao, and Tat-Seng Chua. 2022. KuaiRec: A fully-observed dataset and insights for evaluating recommender systems. In *Proceedings of the 31st ACM International Conference on Information & Knowledge Management (CIKM)*. 540–550.
- [11] Jingtong Gao, Xiangyu Zhao, Muiyang Li, Minghao Zhao, Runze Wu, Ruocheng Guo, Yiding Liu, and Dawei Yin. 2024. SMLP4Rec: An Efficient all-MLP Architecture for Sequential Recommendations. *ACM Transactions on Information Systems* 42, 3 (2024), 1–23.
- [12] Jyotirmoy Gope and Sanjay Kumar Jain. 2017. A survey on solving cold start problem in recommender systems. In *2017 International Conference on Computing, Communication and Automation (ICCCA)*. IEEE, 133–138.
- [13] Jonathan Ho, Ajay Jain, and Pieter Abbeel. 2020. Denoising diffusion probabilistic models. *Advances in neural information processing systems (NeurIPS)* 33 (2020), 6840–6851.
- [14] Feiran Huang, Zefan Wang, Xiao Huang, Yufeng Qian, Zhetao Li, and Hao Chen. 2023. Aligning distillation for cold-start item recommendation. In *Proceedings of the 46th international ACM SIGIR conference on research and development in information retrieval (SIGIR)*. 1147–1157.
- [15] Wang-Cheng Kang and Julian McAuley. 2018. Self-attentive sequential recommendation. In *2018 IEEE international conference on data mining (ICDM)*. IEEE, 197–206.
- [16] Hoyeop Lee, Jinbae Im, Seongwon Jang, Hyunsouk Cho, and Sehee Chung. 2019. Melu: Meta-learned user preference estimator for cold-start recommendation. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (SIGKDD)*. 1073–1082.
- [17] Songyuan Lei, Xinglong Chang, Zhizhi Yu, Dongxiao He, Cuiying Huo, Jianrong Wang, and Di Jin. 2025. Feature-Structure Adaptive Completion Graph Neural Network for Cold-start Recommendation. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, Vol. 39. 12022–12030.
- [18] Jingjing Li, Mengmeng Jing, Ke Lu, Lei Zhu, Yang Yang, and Zi Huang. 2019. From zero-shot learning to cold-start recommendation. In *Proceedings of the AAAI conference on artificial intelligence (AAAI)*, Vol. 33. 4189–4196.
- [19] Zihao Li, Aixin Sun, and Chenliang Li. 2023. Diffurec: A diffusion model for sequential recommendation. *ACM Transactions on Information Systems* 42, 3 (2023), 1–28.
- [20] Xixun Lin, Chuan Zhou, Jia Wu, Lixin Zou, Shirui Pan, Yanan Cao, Bin Wang, Shuaiqiang Wang, and Dawei Yin. 2023. Towards flexible and adaptive neural process for cold-start recommendation. *IEEE Transactions on Knowledge and Data Engineering* (2023).
- [21] Siwei Liu, Iadh Ounis, Craig Macdonald, and Zaiqiao Meng. 2020. A heterogeneous graph neural model for cold-start recommendation. In *Proceedings of the 43rd international ACM SIGIR conference on research and development in information retrieval (SIGIR)*. 2029–2032.
- [22] Xiaoyang Liu, Ziyang Zhang, Xiaoqin Zhang, Pasquale De Meo, and Hocine Cherifi. 2025. Multi-supervisor association network cold start recommendation based on meta-learning. *Expert Systems with Applications* 267 (2025), 126204.
- [23] Fan Lu, Xiaolong Xu, Haolong Xiang, Lianying Qi, Xiaokang Zhou, Fei Dai, and Wanchun Dou. 2025. CLLMRec: Contrastive Learning with LLMs-based View Augmentation for Sequential Recommendation. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence*. 3153–3161.
- [24] Haokai Ma, Ruobing Xie, Lei Meng, Xin Chen, Xu Zhang, Leyu Lin, and Zhanhui Kang. 2024. Plug-in diffusion model for sequential recommendation. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, Vol. 38. 8886–8894.
- [25] Hao Ma, Dengyong Zhou, Chao Liu, Michael R Lyu, and Irwin King. 2011. Recommender systems with social regularization. In *Proceedings of the 4th ACM international conference on Web search and data mining (WSDM)*. 287–296.
- [26] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. 2022. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)*. 10684–10695.
- [27] Yue Shi, Martha Larson, and Alan Hanjalic. 2014. Collaborative filtering beyond the user-item matrix: A survey of the state of the art and future challenges. *Comput. Surveys* 47, 1 (2014), 1–45.
- [28] Min Tang, Shujie Cui, Zhe Jin, Shiuan-ni Liang, Chenliang Li, and Lixin Zou. 2025. Sequential recommendation by reprogramming pretrained transformer. *Information Processing & Management* 62, 1 (2025), 103938.
- [29] Aaron Van den Oord, Sander Dieleman, and Benjamin Schrauwen. 2013. Deep content-based music recommendation. *Advances in neural information processing systems (NeurIPS)* 26 (2013).
- [30] Maksims Volkovs, Guangwei Yu, and Tomi Poutanen. 2017. Dropoutnet: Addressing cold start in recommender systems. *Advances in neural information processing systems (NeurIPS)* 30 (2017).
- [31] Chunyang Wang, Yanmin Zhu, Aixin Sun, Zhaobo Wang, and Ke Wang. 2023. A Preference Learning Decoupling Framework for User Cold-Start Recommendation. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*. 1168–1177.
- [32] Wenjie Wang, Yiyang Xu, Fuli Feng, Xinyu Lin, Xiangnan He, and Tat-Seng Chua. 2023. Diffusion recommender model. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*. 832–841.
- [33] Zihan Wang, Yonghui Yang, Le Wu, Richang Hong, and Meng Wang. 2024. Making Non-overlapping Matters: An Unsupervised Alignment enhanced Cross-Domain Cold-Start Recommendation. *IEEE Transactions on Knowledge and Data Engineering* (2024).
- [34] Le Wu, Junwei Li, Peijie Sun, Richang Hong, Yong Ge, and Meng Wang. 2020. Diffnet++: A neural influence and interest diffusion network for social recommendation. *IEEE Transactions on Knowledge and Data Engineering* 34, 10 (2020), 4753–4766.
- [35] Le Wu, Peijie Sun, Yanjie Fu, Richang Hong, Xiting Wang, and Meng Wang. 2019. A neural influence diffusion model for social recommendation. In *Proceedings of the 42nd international ACM SIGIR conference on research and development in information retrieval (SIGIR)*. 235–244.
- [36] Zihao Wu, Xin Wang, Hong Chen, Kaidong Li, Yi Han, Lifeng Sun, and Wenwu Zhu. 2023. Diff4rec: Sequential recommendation with curriculum-scheduled diffusion augmentation. In *Proceedings of the 31st ACM International Conference on Multimedia (ACM MM)*. 9329–9335.
- [37] Xu Xie, Fei Sun, Zhaoyang Liu, Shiwen Wu, Jinyang Gao, Jiandong Zhang, Bolin Ding, and Bin Cui. 2022. Contrastive learning for sequential recommendation. In *2022 IEEE 38th international conference on data engineering (ICDE)*. IEEE, 1259–1273.
- [38] Xiaolong Xu, Hongsheng Dong, Lianying Qi, Xuyun Zhang, Haolong Xiang, Xiaoyu Xia, Yanwei Xu, and Wanchun Dou. 2024. CmcRec: Cross-modal contrastive learning for user cold-start sequential recommendation. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*. 1589–1598.
- [39] Xiaolong Xu, Hongsheng Dong, Haolong Xiang, Xiyuan Hu, Xiaoyong Li, Xiaoyu Xia, Xuyun Zhang, Lianying Qi, and Wanchun Dou. 2025. C2Rec: causal contrastive learning for user cold-start recommendation with social variable. *ACM Transactions on Information Systems* (2025).
- [40] Yuanbo Xu, En Wang, Yongjian Yang, and Hui Xiong. 2023. GS²-RS: A Generative Approach for Alleviating Cold start and Filter bubbles in Recommender Systems. *IEEE Transactions on Knowledge and Data Engineering* (2023).
- [41] Ling Yang, Zhilong Zhang, Yang Song, Shenda Hong, Runsheng Xu, Yue Zhao, Wentao Zhang, Bin Cui, and Ming-Hsuan Yang. 2023. Diffusion models: A comprehensive survey of methods and applications. *Comput. Surveys* 56, 4 (2023), 1–39.
- [42] Zhengyi Yang, Jiancan Wu, Zhicai Wang, Xiang Wang, Yancheng Yuan, and Xiangnan He. 2024. Generate what you prefer: Reshaping sequential recommendation via guided diffusion. *Advances in Neural Information Processing Systems (NeurIPS)* 36 (2024).
- [43] Ziwei Zhu, Shahin Sefati, Parsa Saadatpanah, and James Caverlee. 2020. Recommendation for new users and new items via randomized training and mixture-of-experts transformation. In *Proceedings of the 43rd International ACM SIGIR*

- Conference on Research and Development in Information Retrieval (SIGIR)*. 1121–1130.
- [44] Xingchen Zou, Yibo Yan, Xixuan Hao, Yuehong Hu, Haomin Wen, Erdong Liu, Junbo Zhang, Yong Li, Tianrui Li, Yu Zheng, et al. 2025. Deep learning for cross-domain data fusion in urban computing: Taxonomy, advances, and outlook. *Information Fusion* 113 (2025), 102606.